

56 collado

December 1987

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---F. G. A. M. Smit has submitted some very interesting quotations on Siphonaptera by early authors. These will be included in installments; the first of these is in this issue.

---lists of additional literature for 1981 (list 13), 1982 (list 10), 1984 (list 8), 1985 (list 6), 1986 (list 4), and 1987 (list 2)

---additions and changes to the mailing list

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15-IX-1901 24-VIII-1987

During his professional career, Dr. Gil Collado published over 100 scientific papers in the fields of entomology and acarology. In the latter specialty, he described three new species of ticks, Rhipicephalus pusillus (1934), Ixodes ventalloi (1936) and Haemaphysalis hispanica (1938), all of which are currently valid. He also recently had an argasid tick named after him: Argas (Persicargas) gilcolladoi Estrada Peña, Lucientes & Sanchez, 1987. In the Siphonaptera, he described the following taxa:

Odontopsylla quirosi (1934)
Araeopsylla hispanica (1934) [=Rhinolophopsylla u. unipectinata]
Chaetopsylla trichosa hispanica (1948) [=Chaetopsylla t. trichosa]
Monopsyllus sciurorum bolivari (1948) [=Ceratophyllus (M.) s. sciurorum]
Echidnophaga tenerifensis (1982)

The latter taxon was described in cooperation with Rodriguez Rodriguez and Zapatero Ramos. He also authored or was coauthor of more major works dealing with the Syrphidae of Spain, malaria, rats and their ectoparasites in the Barcelona area, insect and rodent control, and ectoparasites of domestic animals.

During his life, Dr. Gil Collado received many honors and occupied a number of professional and honorary posts, both nationally and internationally. He was an excellent teacher and taught a number of courses in entomology dealing with

This newsletter is not deemed to be a valid publication for formal taxonomic purposes.

insects and other arthropods of medical importance. He was appreciated and respected by both his students and his colleagues, who admired him for his combination of joviality, affability and professionalism.

(This obituary is a modification of one sent by Fdo. M^a Soledad Gómez, Department of Animal Parasitology, Universidad de Barcelona, 08028 Barcelona, Spain. The editors thank M^a Gómez for supplying us with this, plus additional information about Dr. Gil Collado.)

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CATALOGUE NEWS

The long awaited volume on the fleas of China has finally been received. According to our bibliographers, following is the proper citation:

Liu, Zhiying (Liu C.-y.) (Editor) 1986

Chung-kuo tung wu chih. K'un ch'ung kang. Tsao mu / Chung-kuo k'o hsüeh yüan, Chung-kuo tung wu chih pien chi wei yüan hui.

(Fauna Sinica. Insecta. Siphonaptera. Published by the Editorial Committee of Fauna Sinica, Academia Sinica Science Press. Beijing, China.)

i-xxiii, 1334 pp., 1948 figs.

Following is a list of contributors and their institutional affiliations.

Academy of Military Medical Sciences, Beijing: Liu Zhiying (editor), Wu Houyong, Liu Quan, Ren Qiyu
Guiyang Medical College, Guiyang: Li Guizhen, Huang Guipin
Fujian Medical College, Fuzhou: Wang Dunqing, Liu Zuhua
Yunnan Provincial Institute of Epidemiology and Control: Xie Baoqi, Gong Zhengda

The first 133 pages are devoted to introductory information about fleas, their external and internal anatomy, life cycle and techniques of collection and processing. The following 1091 pages are devoted to the systematics of the Chinese fauna and include illustrations of the diagnostic characters, as well as keys to the taxa, collection records and host and distribution data, all in Chinese except for scientific names and synonymies. Pages 1227 to 1238 contain a brief resumé of the contents of the book in English, followed by an English summary of the 17 new species and subspecies with references to the appropriate text figures. (These taxa are included in the species list and will not be repeated here.) This section is followed by 5 appendices, an extensive bibliography and indices in both Chinese and the Latin alphabet. Pages 1332-1334 list 23 additional taxa that have been described between 1984 and 1986, or otherwise omitted from the text.

This book contains a considerable amount of synonymy, not all of which has been available to non-Chinese workers until now. For the sake of those readers who are taxonomists, following is a resumé of these changes:

adalis = Leptopsylla (Pectinoctenus) pectiniceps pectiniceps
 asiatica = Oropsylla (Oropsylla) alaskensis
 asiaticus = Euhoplopsyllus glacialis profugus
 bijiangensis = Xenodaeria telios
 botis = Amphipsylla primaris mitis
 cansa = Frontopsylla (Frontopsylla) spadix spadix
 carenis = Chaetopsylla (Chaetopsylla) homoea
 certa = Amphipsylla schelkovnikovi
 convexa = Neopsylla abagatui
 crassus = Oropsylla (Oropsylla) silantiewi
 dilatata = Amphipsylla dumalis
 dzetysuensis = Citellophilus altaicus
 elana = Oropsylla (Oropsylla) alaskensis
 fallax = Coptopsylla lamellifer ardua
 geniculata = Vermipsylla alakurt
 gigantica = Vermipsylla alakurt
 gorkha = Frontopsylla (Frontopsylla) diqingensis
 kobayashii = Rhadinopsylla (Actenophthalmus) valenti
 lagrangei = Tunga (Brevidigita) caecigena
 liui = Ischnopsyllus (Ischnopsyllus) liae
 longiloba = Peromyscopsylla ostsibirica
 manchurica = Peromyscopsylla ostsibirica (listed in the index but not in
 synonymy, p. 689. Probably available.)
 musculi = Leptopsylla (Leptopsylla) segnis
 nana = Amphipsylla casis
 passerinus = Ceratophyllus (Ceratophyllus) gallinae tribulus?
 rhombosa = Neopsylla honora
 segregatus = Oropsylla (Oropsylla) silantiewi
 solitaria = Stenoponia conspecta
 stejnegeri = Oropsylla (Oropsylla) alaskensis (listed in index but not
 in synonymy, p. 1081)
 subcaecatus = Paradoxopsyllus curvispinus
 subula = Ctenophyllus [=Ochotonobius] hirticrus
 tamias = Ceratophyllus (Monopsyllus) indages
 tateishii = Ischnopsyllus (Hexactenopsylla) indicus
 tenuihamus = Citellophilus [=Callopsylla] sparsilis sparsilis
 terribilis = Ctenophyllus armatus
 transbaikalicus = Paradoxopsyllus scorodumovi
 transbaikalica = Amphipsylla primaris mitis
 utia = Frontopsylla (Orfrontia) frontalis baikal
 weisskei = Oropsylla (Oropsylla) silantiewi
 yunnanicus = Macrostylophora hastata menghaiensis

This volume should prove useful, even to workers who do not read Chinese,
 since it is amply illustrated with drawings of average or higher quality.
 Finally our Chinese colleagues have a firm basis from which to extend their
 future field studies. Congratulations to the authors are well deserved.

We are informed by FRANS SMIT that VOLUME VII of the Catalogue of the Roth-
 schild Collection of Fleas in the British Museum (Natural History) has been re-
 leased by the Oxford University Press. According to him it consists of 380 pages,
 710 figures and 5 plates with another 38 figures, dealing with the families

Malacopsyllidae and Rhopalopsyllidae. Although Mr. Smit has apprised us of the taxonomic changes contained in the volume, we have not seen it yet. Accordingly, we will wait to write a review of it until we have had the opportunity to examine the volume.

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LISTS AND LISTS

For those recipients of Flea News who have tried to maintain their copy of Smit & Wright, 1978, A list of code numbers of species and subspecies of Siphonaptera, attention is called to the following additions:

96430 geniculata Li Vermipsylla geniculata = 88501
96431 gigantea Li Vermipsylla geniculata = 88501

Also, omitted from the list for 1985 was the following:

98518 lobatschevi Cyprich, Kiefer & Krumpál Citellophilus sungaris

While on the subject of lists, in addition to a current version of the Smit & Wright, 1978, list, we maintain the following in our computer:

Alphabetical list of valid species group names
in the order Siphonaptera (47 p.)

Following is an alphabetically arranged list of all of the species group names in the order Siphonaptera that are currently deemed valid. This list is maintained for use in our laboratory and does not constitute a publication, nor should it be so cited. Although it has been proofed it is certain to contain additional errors. When noted, these should be called to the attention of R. E. Lewis at the address given on the title page.

ALPHABETICAL LIST OF SIPHONAPTERA SPECIES

A

92221 abacetus (J. & R., 1922) Orthopsylloides
94667 abagaitui Ioff, 1946 Neopsylla
90508 abantis (Rothschild, 1905) Megabothris (Ameg)
93032 abaris Jordan, 1930 Dinopsyllus (Dino)
98106 abazhouensis Liu, Liu & Zhai, 1981 Macrostylophora
94624 abramovi (Ioff, 1946) Nosopsyllus (Gerb)
90503 acamantis acamantis (Rothschild, 1905) Oropsylla (Thra)
92532 acamantis howelli (Jordan, 1925) Oropsylla (Thra)
97001 acamantis medius (Stark, 1970) Oropsylla (Thra)
95717 acamantis pristinus (Stark, 1957) Oropsylla (Thra)
93604 acamantis utahensis (Wagner, 1936) Oropsylla (Thra)
92309 acanthina J. & R., 1923 Neopsylla
91318 acanthurus J. & R., 1913 Ctenophthalmus (Ethi)
97408 acanthus Lewis, 1974 Paradoxopsyllus
93004 accola Wagner, 1930 Rhadinopsylla (Acte)

Sample

Sample

and

Siphonaptera genus and species list,
including code numbers 1758-1986 (110 p.)

Following is a list of the recognized valid taxa belonging to the order Siphonaptera. They are arranged in alphabetical order, first by genus and then by species. Entries include the following information: code number, species, subspecies (where appropriate), author, and date of publication. Following this information is usually a number which corresponds to the number of the reprint in my file carrying the original description. If a translation of a foreign paper is available, this is usually indicated. After some entries another set of numbers may appear in parens, i. e. (1/1). These are species in my collection and the numbers indicate the numbers of male/female specimens. Where large numbers are available the word (many) may be substituted for numbers. Information concerning the content of my collection may be obtained by writing me at the address on the cover page.

ACANTHOPSYLLA J. & R. 1922 (22)

96907	aiyurensis Holland, 1969	B	(1/1)
95356	bella Smit, 1953	16	
96910	bisinuata Holland, 1969	B	
95355	divergens Smit, 1953	16	
97105	dunneti Holland, 1971	2907	
93315	enderleini (Wagner, 1933)	4122	
96909	eudromiciae Holland, 1969	B	(1/1)
93622	franklinensis M. Rothschild, 1936	1778	

With the advent of computers it has been relatively simple to keep these lists current without the burden of retyping and proof reading. Now we are considering making these lists available to interested parties for cost of production and postage. Currently, printing costs are approximately 5¢ per page. At this rate both lists would cost US\$7.85 plus some type of binding and postage. If you are interested in copies, please inform us in writing. Our course of action will be determined by your response.

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NOTES

Readers having business with the Siphonaptera Section of the British Museum (Natural History) should now address their enquiries to Mrs. Annette K. Seccombe.

ECTOPARASITES FOR SALE. Frans Smit has informed us that he has 2 complete sets of Ectoparasites that he would like to sell. This rare series (only 200 copies were printed) consists of 6 numbers published from December 1915 through June 1924. It contains articles on the Polycetenidae (2), Clinocoridae [= Cimicidae] (2) and Siphonaptera (27) by K. Jordan and/or N. C. Rothschild, and is of particular interest to pulicologists because of the large number of descriptions of new species. Since most copies of this work are tied up in institutional libraries and thus not available on the open market, Mr. Smit's offer to sell presents a rare opportunity. He may be contacted at 4, Glamis Gardens, Longthorpe, Peterborough PE3 6PQ, England.

PLAGUE UPDATE. According to the October Newsletter of the Society of Vector Ecologists, the incidence of plague is on the decline in the United States. According to Dr. Alan Barnes of the Communicable Disease Center, Fort Collins, Colorado, human cases of the disease have dropped from a high of 40 in 1983 and 31 in 1984, to 17 in 1985, 11 in 1986, and 7 so far in 1987, including one fatality in Oregon. Barnes goes on to predict an increase in plague activity from the 1987 low.

MURINE TYPHUS OUTBREAK. Brazil - A recently received report indicates that a case of murine (endemic, flea-borne) typhus occurred in Porto Alegre, Rio Grande do Sul, in February. Murine typhus has not been reported from that area in at least 10 years, and the state epidemiology service reportedly was unfamiliar with the disease.

---T.I.B. 8/9 87

FINANCIAL WOES FOR TICK LAB. A recent article in the Washington Post carried the banner "Budget Woes May Close National Tick Lab." This laboratory, which is operated by the National Institute of Allergy and Infectious Diseases of the National Institutes of Health (NIH), is the only one in the world where ticks can be identified on a global basis and where information can be obtained on the habits and life histories of these highly important vectors of disease.

The article went on to point out that the tick collection, moved to Washington only three years ago from the Rocky Mountain Laboratories in Hamilton, Montana, requires \$140,000 a year in support. This is the largest tick collection in the world and contains about 90% of the approximately 850 described taxa, and includes the renowned collection and library of the late Harry Hoogstraal. NIH Deputy Director Joseph E. Rall is quoted as having said: "While we agree ... that research on the classification of ticks is ... important ..., the No. 1 ... priority identified by the Department of Health and Human Services is AIDS."

The tick unit and collection is currently housed in the Smithsonian Institution's Museum Support Center in Suitland, Maryland, under the direction of Dr. James E. Keirans. This unit is scheduled to be closed September 30, 1988, unless the authorities come to realize that termination of this unique facility is a prime example of false economy. Research on AIDS is surely important enough in its own right for NIH to receive sufficient funds to handle the problem without sacrificing this critically needed laboratory in order to effect the savings of two personnel slots and \$140,000.

The current plan to close this laboratory is another example of the lack of appreciation by administrative and budgetary officials of the importance of the study of systematics in general and vectors of diseases in particular. In recent years, as specialists in systematics of fleas, ticks, lice, trombiculid mites, and mosquitoes have retired or died, they have not been replaced in museums or governmental laboratories throughout the world.

Also, there has been a serious misunderstanding about the effectiveness of the fund-raising campaign launched by the Smithsonian Institution to obtain private funding for the establishment of the Hoogstraal Center for Tick Research. Thus, the Fall issue of the Bulletin of the Entomological Society of America reported that \$1.5 million had been raised from private sources for this tick research center. Actually, only about \$30,000 has been received. The \$1.5 million figure that was quoted represents the goal of the campaign.

---R. Traub

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QUOTATIONS FROM SOME EARLY AUTHORS

A thorough history of the study of fleas has still to be written. This will not be an easy task as, e.g., it may be quite difficult to locate and consult a number of the early publications. Over the years I made some extracts from old sources which I now offer for amusement, perhaps thereby whetting someone's appetite.

---F. G. A. M. Smit

DEFINITION

Fleas are black, round, animals which suck blood with a rostrum which causes swellings; apart from six walking legs they have long jumping legs. (Magnus 1255-1270)

It [the flea] seems as big as a little Prawn or Shrimp, ... (Power 1664:1) His head, body, and limbs also, be all of blackish armour-work, shining and polished with jemmar's, most excellently contrived for the nimble motion of all the parts: nature having armed him thus Cap-a-pe [cap-à-pie = from head to foot] like a Curiazier in warr, that he might not be hurt by the great leaps he takes; ... His neck, body, and limbs are also beset with hairs and bristles, like so many Turnpikes, as if his armour was palysado'd about by them. (Power 1664: 1-2)

One would wonder at the great strength lodged in so small a Receptable, and that he is not able onely to carry his whole about him, but will frisk and curvet so nimbly with it: Stick a large brass pin through his tayl and he will readily drag it away. (Power 1664: 2-3)

COLOUR

They are black and shining, their breast and belly is yellow from black, in white Dogs they are more clear, in red more yellow, in black Dogs blacker than in others. (Moufet 1634: 275; 1658: 1101)

Fleas are tiny black vermin ... (Cantimpré 1233-1248)

REFERENCES

- ALDROVANDUS, U. 1602 De animalibus insectis, Historiae de insectis. Bononiae. [8+] 767 pp., [+ index], figs. (Siph.: 563-569)
- ANGLICUS, Bartholomaeus 1483 De Proprietatibus rerum. Nuremberg. (Siph.: Liber XVIII, Capitulum LXXXVII)
- ANONYMOUS 1392-1394 Menagier de Paris. [translation by E. Power, 1928, London]
- ANONYMOUS 1491 Ortus sanitatis. Mainz. 453 leaves in double columns, 1073 figs. (Siph.: Capitulum CXVIII, 1 fig.)
- ARISTOTELES +300 B.C. Historia animalium.
- BOUTON, J. 1640 Relation de l'establissement des Francois depuis l'an 1635 l'isle de la Martinique, l'une des antilles de l'Amerique. Paris. [6+] 141 pp. (Siph.: 91-92)
- CANTIMPRÉ, T. 1233-1248 Liber de natura rerum.
- CHARLETON, G. 1677 Exercitationes de differentiis & nominibus animalium, Quibus accedunt mantissa anatomica, et quaedam de variis fossilium generibus. (ed. 2) Oxoniae. [18+] 119 + 106 + 78 pp. [+ index]. (Siph.: (1) 53)
- COLUMELLA, L.J.M. +50 De re rustica.
- GRANDI, L. 1667 Alfabetto di secreti medicinali, et altri curiosi, e diletteuoli d'ogni materia, con l'arte facile d'uccellare, e pescare. Bolagna. 216 pp. (Siph.: 146)
- HAKLUYT, R. 1600 The third and last volume of the voyages, navigations, traffiques, and discoveries of the English Nation, etc. London. [14+] 868 pp. (Siph.: 449)
- HERRERA, A. DE 1706 Vierde en laatste scheeps-togt door Christoffel Kolumbus etc. naar de West-Indien. In: Gottfried, J.L. - De gedenkwaardige en alom beroemde voyagien der Spanjaarden na West-Indiën, etc. Leyden. I. 56 columns. (Siph.: 34)
- HILDEGARD [Hildegard von Bingen] 1533 Physica S. Hildegardis elementorum, etc. (Siph.: 118-119)
- JONSTON, J. 1653 Historiae naturalis de insectis. Libri III. Francofurti ad Moenum. 200 pp., 26 pls. (Siph.: 123-124, 140-141)
- LEO, J. 1550 Delle descrittione dell'Africa. In: G.B. Ramusio - Delle navigationi et viaggi. Venetia. 1: fol. 1-95. (Siph.: 14F, 27C, 74A, 86A)
- LEO, J. 1600 A geographical historie of Africa, written in Arabicke and Italian (translation by John Pory). Londini. VI+420 pp. (Siph.: 56, 107, 271, 315)

- LÉRY, J. DE 1593 *Schiffart in Brasilien in America, etc.* Frankfurt am Mayn. [28+] 93+285 pp., figs. (Siph.: 166-167)
- LÉRY, J. DE 1594 *Histoire d'un voyage fait en la terre du Bresil, autrement dite Amerique.* [42+] 382 pp. [+index], 1 pl. (Siph.: 160-162)
- MAERLANT, J. VAN 1265-1269 *Der naturen bloeme.* [translation of A. Magnus 1255-1270]
- MAGNUS, A. [A. VON BOLLSTÄDT] 1255-1270 *De animalibus.*
- MAGNUS, O. 1565 *Historia delle genti et della natura della cose settentrionali.* Venetia. 286 fol., figs. (Siph.: fol. 226)
- MOUFET, T. 1634 *Insectorum sive minimorum Animalium Theatrum.* London. XX+326 pp. (Siph.: 268, 275-278)
- MOUFET, T. 1658 *The Theater of insects: or lesser living creatures.* In: *Topsel - The History of four-footed beasts and serpents; etc.* London. pp. 889-1130, figs. (Siph.: 1096, 1101-1103)
- OVIEDO Y VALDÉS, G.F. DE 1535 *La historia general de las Indias.* Sevilla. CXCIII fol. (Siph.: fol. XXI)
- PALLADIUS, R.J.A. +380 *De re rustica.* Libri XIV.
- PAPYROS EBERS +1550 B.C. [see: G. Ebers & L. Stern, 1875. *Papyros Ebers, das hermeutische Buch Über die Arzneimittel der alten Aegypter in hieratischer Schrift.* Leipzig. 2 vols.]
- PLINIUS Secundus Maior, C. 77 *Historia naturalis.*
- POWER, H. 1664 *Experimental Philosophy, In three Books: Containing new experiments - Microscopical, Mercurial, Magnetical, With some Deductions, and probable Hypotheses, raised from them, in Avouchment and Illustration of the now famous Atomical Hypothesis.* London. [20+] 193 pp. (Siph.: 1-3)
- PURCHAS, S. 1625 *His pilgrimes in five bookes.* London. 2: 749-1860 (Siph.: 781, 824, 840); 3: 841-1140 (Siph.: 975); 4: 1141-1973 (Siph.: 1252, 1365)
- SCHMIDEL, U. 1599 *Warhafftige Historien. Einer wunderbaren Schiffart, etc.* Noribergae. [9+] 103 pp., figs. (Siph.: 85)
- SEVERINUS, M.A. 1645 *Zootomia democritaea.* Noribergae. [16+] 408 [+34] pp. (Siph.: 246, 251)
- STADEN, J. 1557 *Wahrhaftige history, etc.* (Siph.: Chapter 33)
- STADEN, J. 1593 *Brasilia.* Frankfurt am Mayn. [14+] 92 pp., figs. (Siph.: 89)
- TALMUD, 200-500 *Horajoth* 11, a; *Shabbath* 12, a, 107, b; *Sch. Aruch Jore Dea*, paragraph 179, 5; *Rabba Wajikra*, fol. 189, col. 3.
- THEVET, F.A. 1558 *Les singularitez de la France antarctique, autrement nommée Amerique, & de plusieurs Terres & Isles decouvertes de nostre temps.* Anvers. 166 pp., figs. (Siph.: 88)
- VRIES, S. DE 1682 *Curieuse aenmerckingen der bysonderste Oost en West Indische verwonderens-waerdige Dingen; nevens die van China, Africa, en andere gewesten des werelds.* Utrecht. I: 1-496 (Siph.: 329-330); II: 497-1328 (Siph.: 500, 501, 580)

[The references to all the quotations are being included in the first installment so that readers may have the complete citations as successive installments are published. ---R.E.L. & J.H.L.]

REPRINTS RECEIVED - We wish to thank the following for supplying reprints and citations since the publication of Flea News 33:

A. Alekseev	L. Durden	J. Keirans	J. Ryba
J. Beaucournu	R. George	O. Larson	W. Skuratowicz
V. Cerny	P. Gilbert	A. Main	D. Sleeman
D. Cyprich	J. Greve	S. Medvedev	F. Smit
A. Dudich	R. Haitlinger		

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PAPERS PRESENTED - At the Annual Meeting of the Entomological Society of America, Boston, Massachusetts, November 29-December 3, 1987, the following papers were presented:

Rust, M. K. & D. A. Reiersen

Effect of IGRs on cat fleas: Oral versus contact exposure.

Lemke, L. A.

The development of new IGRs for cat flea control: Laboratory and field tests.

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MAILING LIST

Additions

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DELETIONS

Dr. M. Portús Vinyeta

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PROVISIONAL LISTS OF TAXA DESCRIBED IN 1986 AND 1987

1986

Coorilla allisoni Mardon
Ischnopsyllus (Hexactenopsylla) ussuriensis Medvedev
Frontopsylla (Frontopsylla) wagneri superjecta Zhang, Liu & Zhang
Paradoxopsyllus jinshajiangensis concavus Liu, Liu & Zhang
Lentistivalius affinis Li
Plocopsylla diana Beaucournu, Gallardo & Launay
Neopsylla affinis deqinensis Xie & Li
Ctenophthalmus (Ctenophthalmus) bifidatus mancus Orhan & Beaucournu
Ctenophthalmus (Mediostenophthalmus) nifetodes tvrtkovici Brelih
Ctenophthalmus (Mediostenophthalmus) nifetodes petrovi Brelih
Ctenophthalmus (Mediostenophthalmus) nifetodes krystufeki Brelih
Ctenophthalmus (Mediostenophthalmus) nifetodes dzukici Brelih
Callopsylla (Orneacus) beishanensis Wu, Ni & Wu
Pariodontis riggenbachi yunnanensis Li & Yan
Nearctopsylla (Neochinopsylla) xijiensis Wu, Chen & Li
Ctenophthalmus (Sinostenophthalmus) proboscis Wu, Xiue & Hu
Ctenophthalmus (Sinostenophthalmus) taiwanus terrestris Chen, Ji & Wu
Peromyscopsylla himalaica sichuanoyunnana Xie, Chen & Liu
Ornithophaga anomala qinghaiensis Liu, Cai & Wu
Geusibia apromina sichuanensis Liu, Zhai & Liu
Frontopsylla (Frontopsylla) spadix borealosichuana Liu & Zhai
Frontopsylla (Frontopsylla) spadix shennongjiaensis Ji, Chen & Liu
Frontopsylla (Frontopsylla) nakagawai qinghaiensis Liu, Cai & Pan
Frontopsylla (Frontopsylla) nakagawai borealosinica Liu, Wu & Chang
Frontopsylla (Frontopsylla) exilidigita tiebuensis Chen, Wei & Liu
Paradoxopsyllus stenotus Liu, Cai & Wu
Paradoxopsyllus diversus Liu, Chen & Liu
Macrostylophora hastata hainanensis Liu & Pan
Nosopsyllus (Nosopsyllus) wualis leizhouensis Li, Huang & Liu
Ceratophyllus (Monopsyllus) fengi Liu, Xie & Wang

1987

Caenopsylla janineae Beaucournu & Gouat
Amphipsylla longispina gongheensis Zhang & Ma
Plocopsylla viracocha Schramm & Lewis

(1981 - 1984)

Plocopsylla nungui Schramm & Lewis
Plocopsylla athena Schramm & Lewis
Plocopsylla kilya Schramm & Lewis

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Answer to FLEA TRIVIA QUESTION in Flea News 34: 1918 was the only year in this century (so far) in which no new flea species were described.

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LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 13)

Lim Boo Liat, N. Sustriayu, I. Ladera & Y. H. Bang - Field trials of fenitrothion, malathion, and DDT dusts against fleas on *Rattus rattus diardii* in Ciloto, West Java, Indonesia. WHO/VBC/81.802.

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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

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The editors of *Flea News* extend to all our readers our very best wishes for a happy and prosperous 1988.